

Modeling the Sick Leave Decision: Constraint vs. Cognition

Anita Tsai

Professor: Michael Hayashi

Honors Cognitive Science Thesis

4/15/2026

Abstract:

Access to paid sick leave in the United States remains uneven, leading many workers to continue working while ill despite potential health risks. While prior research has examined the structural and policy determinants of sick leave utilization, less attention has been given to the cognitive processes underlying individual decision-making in this context. This study develops a theoretical, decision-theoretic model of sick leave behavior by integrating constructs from the Health Belief Model (HBM) within a utility-based framework.

The model conceptualizes the decision to take sick leave as the result of combining perceived susceptibility, severity, benefits, barriers, self-efficacy, and cues to action into an overall utility function. Using scenario-based analysis, the study examines how variations in illness type, worker characteristics (e.g., hourly vs. salaried), and functional representations of beliefs influence predicted behavior. Functional forms are chosen to reflect different cognitive processing patterns, including proportional, threshold-based, and nonlinear responses.

Results from the modeling framework suggest that structural constraints, particularly perceived barriers, often dominate decision outcomes, while cognitive factors such as legitimacy thresholds and self-efficacy shape how illness signals are translated into action. These findings highlight that working while ill may represent a rational response to perceived constraints rather than a failure of judgment. Overall, this study demonstrates the value of combining cognitive perspectives to better understand sick leave decisions, and provides a foundation for future work incorporating empirical validation and more complex behavioral interactions.

Modeling the Sick Leave Decision: Constraint vs. Cognition

Introduction:

Access to paid sick leave in the United States is highly uneven, shaped by employment type, industry, and state-level legislation rather than a universal federal mandate (Bureau of Labor Statistics [BLS], 2023; Kaiser Family Foundation [KFF], 2020). As a result, workers in sectors such as retail, hospitality, food service, and manufacturing, many of whom are hourly, non-unionized, and required to work onsite, are significantly less likely to have access to paid sick leave than workers in other sectors (BLS, 2023). For these workers, taking time off when ill often results in immediate income loss and, in some cases, fear of job insecurity or retaliation (Schneider & Harknett, 2021).

These structural gaps have consequences for both individual and public health. Workers without access to paid sick leave are more likely to work while ill, a phenomenon known as presenteeism (Kumar et al., 2012; Pichler & Ziebarth, 2017). Presenteeism has been linked to prolonged illness, reduced productivity, and increased transmission of infectious diseases in workplace settings (Hemp, 2004; Kumar et al., 2012). During public health crises such as the COVID-19 pandemic, these effects became especially visible, as workers in high-contact, low-wage occupations faced strong incentives to remain at work despite symptoms, contributing to workplace outbreaks and broader community spread (Centers for Disease Control and Prevention [CDC], 2020; Schneider & Harknett, 2021).

Existing research in public health, labor economics, and social policy demonstrates that access to paid sick leave is associated with reduced presenteeism, lower disease transmission, and improved worker well-being (Kumar et al., 2012; Pichler & Ziebarth, 2017). State-mandated sick leave policies in the U.S. increase leave utilization and reduce workplace illness (Pichler & Ziebarth, 2017), while European countries with comprehensive sick leave systems report lower rates of working while sick (Eurofound, 2012; OECD, 2019). However, much of this literature

focuses on policy presence and aggregate outcomes like infection rates or labor supply, rather than on the cognitive mechanisms that generate individual sick leave decisions.

At the individual level, the recognition of illness does not lead to a single predictable outcome. Behavioral economics demonstrates that individuals discount delayed benefits, overweight immediate costs, and rely on heuristics when navigating uncertainty (Camerer et al., 2004; Loewenstein et al., 2001), while Bounded Rationality emphasizes the role of environmental constraints and perceived feasibility in shaping decisions (Simon, 1955). As a result, an employee may recognize that their illness is clinically significant yet still choose to work due to financial pressure, workplace norms, or perceived lack of legitimacy. Consistent with this view, Karanika-Murray and Biron (2020) argue that presenteeism often reflects a strategic and adaptive response to competing constraints, rather than a failure of judgment. Furthermore, behavioral research further highlights that health-related decisions are shaped not only by objective risk, but also by subjective perceptions of vulnerability, social consequences, and personal goals (Merrell et al., 2024). Together, these perspectives suggest that working while ill may represent a rational response to perceived barriers, uncertainty, and competing incentives.

Despite these insights, there remains limited empirical work examining sick leave decisions at the individual level. Less attention has been paid to how the same construct may operate differently across illness contexts and worker types. Mathematical modeling provides a useful framework for addressing this gap by allowing researchers to formally represent how multiple factors are cognitively integrated into action under constraint. Accordingly, understanding sick leave utilization requires a shift from analyzing policy presence to modeling the decision-making process itself.

To formalize this process, this study adopts the Health Belief Model (HBM) as its theoretical foundation (Rosenstock, 1974). The HBM defines six key constructs, discussed in depth within the methodology section, involved in health behavior choices: perceived

susceptibility, perceived severity, perceived benefits, perceived barriers, self-efficacy, and cues to action (Becker, 1974; Champion & Skinner, 2008). However, most applications of the HBM rely on empirical, correlational approaches, using survey or observational data to examine whether these constructs predict observed behavior (Champion & Skinner, 2008). While such studies provide evidence that HBM components are associated with health-related actions, they do not typically model how individuals cognitively integrate these beliefs into a decision process or evaluate whether a formalized model accurately reproduces behavioral outcomes. As a result, the HBM is most often used descriptively, rather than as a fully specified decision-making framework.

This thesis addresses that gap by designing a decision-theoretic model of sick leave behavior based on the HBM. By operationalizing constructs from the HBM as terms in a utility function, this work provides a theoretical framework to evaluate sick leave decisions as the result of an individual's consideration and integration of multiple factors. I use scenario analysis to consider the impact of illness type, age, and compensation structure on sick leave decisions. Scenarios capture a range of hypothetical configurations of illness and individual behavior.

Methods

Applying Decision Theory to Operationalize Health Belief Model

The goal of this study is to examine how these beliefs may be cognitively integrated within a decision-theoretic framework. To do that, I developed a theoretical, simulation-based model of sick leave decision-making. The model is used to conduct scenario-based analyses, exploring how variations in illness type, worker characteristics, and structural constraints influence decision outcomes. The emphasis is not on reproducing real-world behavioral data, but on understanding how different modeling assumptions shape predicted behavior.

More specifically, this study conceptualizes the choice to take sick leave as the outcome of a utility-based decision-making process. The model adopts an additive utility structure, in which each construct contributes independently to overall utility. This reflects the simplifying

assumption that individuals evaluate perceived susceptibility, severity, benefits, and other factors separately, rather than through interacting or multiplicative processes. While interaction terms (e.g., susceptibility $\times$ severity) could capture more complex dependencies, incorporating such nonlinear relationships would introduce additional assumptions and substantially increase model complexity. The present model therefore prioritizes interpretability by isolating the independent contribution of each construct. Weights are assumed to be uniform across constructs, and are implicitly incorporated within the coefficients of each functional form. This approach allows the model to focus on how different functional transformations shape behavior, rather than estimating empirically derived weights.

The following model implements HBM constructs within a rational choice framework in which individuals are assumed to select the option that maximizes their perceived best interest. The utility of taking sick leave is defined as the sum of the contributions of each HBM construct:

$$U(\textit{sick leave}) = u(SU) + u(SV) + u(BE) + u(SE) + u(Cue) + u(UR)$$

Within the HBM model, *SU* represents perceived susceptibility, *SV* represents perceived severity, *BE* represents perceived benefits of taking leave, *SE* represents self-efficacy, *CUE* represents cues to action, and *UR* represents perceived barriers. Barriers are represented as negative values, reflecting the perceived costs of absence. All belief inputs are scaled on a continuous range from 0 to 1. This normalization ensures consistent comparison across constructs and illness contexts. For simplicity I assume that the utility of not taking sick leave is 0. When the utility of taking sick leave exceeds that of continuing to work ($U(\textit{sick leave}) > 0$), the model predicts that the individual remains home; when it does not, the model predicts continued work attendance. Importantly, this model relies on a set of simplifying assumptions to make the decision process tractable. Alternative model specifications such as having nonlinear interaction terms or alternative functional forms may better capture certain aspects of real-world decision-making; however, exploring these possibilities would substantially increase model complexity. Thus, this study prioritizes a more constrained framework to isolate key

mechanisms, leaving more complex formulations as an avenue for future work. Following this, parameter values and coefficients are assigned to produce behaviorally interpretable patterns rather than being empirically estimated. These assumptions form the basis of the model and are not intended to fully capture real-world complexity, but rather to provide a structured framework for examining how different belief configurations influence decision outcomes.

Functional Transformations of Beliefs

Each belief construct is operationalized through a persona-specific utility function. In this study, 'personas' represent distinct, archetypical profiles of workers categorized by key demographics and employment structures (such as young hourly employees versus older salaried employees). These personas are intended to model the cognitive processes of different types of individuals. In particular how they weigh, suppress, or amplify signals based on their specific life and work constraints.

Different math functions are used to capture how beliefs vary in their influence on decision-making across contexts. Linear functions are used when beliefs are processed proportionally, such that increases in perceived intensity of the construct produce corresponding increases in decision weight. For example, an individual may gradually place more value on taking sick leave as symptoms like fatigue or discomfort steadily worsen, without any abrupt shifts or thresholds. In contrast, convex functions are used to represent the assumption that individuals exhibit delayed sensitivity, where low levels of a signal are discounted but higher levels produce disproportionately stronger responses. For example, a worker experiencing mild fatigue or a slight sore throat may continue working because these symptoms are perceived as manageable; however, as symptoms escalate to include high fever or severe exhaustion, the perceived severity rises sharply, leading to a disproportionate increase in the utility of taking sick leave. Sigmoid functions, by comparison, represent threshold-based processing. This is particularly relevant in contexts such as mental health, where individuals may not consider taking leave for mild stress but begin to do so once symptoms reach a level that feels socially or

personally “valid,” or after receiving validation from others. Finally, step functions capture binary triggers, serving as a more dramatic and sharper version of threshold-based processing that produces discrete shifts in behavior when specific conditions are met. For instance, an employee may only take sick leave after receiving explicit managerial approval or a formal medical recommendation. Together, these math functions allow the model to capture variation in the magnitude of beliefs and how and when they become behaviorally consequential.

Perceived Susceptibility

Perceived susceptibility reflects the individual’s belief about the likelihood of experiencing worsening health, prolonged recovery, or recurrence if work continues. A value near 0 indicates the belief that continued work poses little to no risk, whereas a value near 1 indicates strong anticipation of negative health consequences. In the context of a mild acute illness such as a cold, susceptibility captures whether an individual believes that working through symptoms will have minimal impact versus increasing the likelihood of symptom escalation, longer recovery time, or even spreading illness to others. In chronic illness contexts, susceptibility is less about acute worsening and more about expectations of persistence and cumulative strain. For example, an individual with chronic back pain may believe that continuing to work will not cause immediate deterioration but will prolong discomfort, exacerbate inflammation over time, or delay recovery cycles. In acute mental health contexts, susceptibility reflects concern about psychological deterioration or relapse, such as the belief that continuing to work while experiencing stress or burnout may intensify emotional exhaustion, reduce coping capacity, or trigger more severe episodes of anxiety or depression. Functional transformations of susceptibility differ across personas to capture patterns such as normalization of recurring symptoms, heightened vigilance to early warning signs, or delayed recognition of risk, allowing the same underlying belief to influence behavior differently depending on how it is interpreted and weighted (Table 2).

Perceived Severity

Perceived severity represents beliefs about the seriousness of the condition, including anticipated functional impairment and long-term consequences. A value of 0 reflects trivial discomfort, while a value of 1 reflects catastrophic or incapacitating disability. Severity differs from susceptibility in that it concerns current magnitude rather than future risk. In the model, severity may be processed linearly when individuals respond proportionally to symptoms, or through convex or sigmoid functions when legitimacy thresholds or endurance norms delay action (Table 2). These functional differences reflect empirical findings that individuals often discount early symptoms and respond only after crossing perceived thresholds of seriousness.

Perceived Benefits

Perceived benefits represent the expected positive impacts of taking sick leave on an individual's prognosis. A value near 0 indicates the belief that staying home will not meaningfully improve the condition, whereas a value near 1 indicates strong belief in recovery or prevention benefits. Benefits are particularly context-dependent. In mild acute illness, benefits may be moderate due to expected short-term recovery. In chronic illness, benefits may be low because short-term rest does not resolve long-term conditions. Functional forms are chosen to reflect whether individuals perceive rest as preventive investment, temporary relief, or ineffective intervention.

Perceived Barriers

Perceived barriers represent structural, financial, reputational, and logistical costs associated with absence. A value of 1 represents substantial structural constraint, such as meaningful wage loss or employment risk. Values between approximately 0.8 and 0.9 represent significant economic or reputational pressure. A value near 0.5 represents moderate social discomfort or workload accumulation without financial harm. A value approaching 0 represents a supportive environment in which absence carries minimal penalty. Barriers are modeled as a cost term, contributing negative utility. The magnitude of the barrier term reflects the degree to

which structural constraints anchor decision-making. Larger barrier values indicate that external constraints dominate illness signals.

Self-Efficacy

Self-efficacy represents an individual's confidence in their ability to request or take sick leave. A value near 0 reflects low confidence in communicating needs, navigating workplace norms, or anticipating supportive responses, whereas a value near 1 reflects strong perceived agency and autonomy. Within this model, self-efficacy functions as a factor that determines whether illness-related beliefs are translated into action. From a cognitive science perspective, self-efficacy can be understood as influencing action selection and perceived control within decision-making systems: even when perceived severity or susceptibility is high, low self-efficacy may inhibit behavior by increasing anticipated social or procedural costs. For example, an employee experiencing severe symptoms may still choose to work if they lack confidence in how to notify their manager, fear negative evaluation, or are uncertain about workplace policies. Conversely, high self-efficacy can amplify the behavioral impact of illness beliefs by lowering the cognitive and social barriers to action. Functional transformations of self-efficacy therefore capture whether perceived autonomy enhances or attenuates the influence of other belief components, reflecting its role as a gating or amplification mechanism in decision processes.

Cues to Action

Cues to action represent internal or external triggers that prompt an individual to reconsider or change their behavior. A value of 0 reflects the absence of prompting, where the individual continues their current course of action without interruption, while a value of 1 reflects strong, directive cues that actively push the individual toward taking sick leave. These cues can take many forms. Internal cues include recognizing that symptoms are worsening, such as feeling unusually fatigued, dizzy, or unable to concentrate. External cues include interpersonal and organizational signals, such as a conversation with a manager encouraging the use of sick

leave (“You should take the day off and rest”), a coworker expressing concern (“You don’t look well today”), or institutional messaging like workplace posters or emails that say “Please stay home if you are sick.” More formal cues may include medical advice, a doctor’s note, or a positive test result, which can serve as strong validation for taking leave. This term reflects how individuals often rely on external legitimacy signals to justify behavior change, even when internal symptoms are already present.

Function Choice Rationale

I chose each functional form to reflect a specific hypothesis about how individuals cognitively process different types of beliefs. The coefficients within each functional form serve as scaling parameters that shape how strongly each construct influences utility. These values are not intended to represent empirically estimated weights, but rather to produce function shapes that align with qualitative hypotheses about behavior. For instance, smaller coefficients reflect attenuated influence (e.g., when individuals discount certain signals), while larger coefficients reflect stronger responsiveness. In this sense, coefficients can be interpreted as encoding relative importance, while also ensuring that the functional form exhibits the intended behavioral pattern.

	Mild Cold	Mental Health	Chronic Health
Young Hourly	$u_i(SU) = 0.5\sqrt{(SU)}$	$u_i(SU) = 0.1 \times SU$	$u_i(SU) = 1/(1 + e^{(-8(SU-0.8))})$
	$u_i(SV) = 1/(1 + e^{(-15(SV-0.7))})$	$u_i(SV) = 1/(1 + e^{(-15(SV-0.95))})$	$u_i(SV) = 1/(1 + e^{(-15(SV-0.9))})$
	$u_i(BE) = 0.3\sqrt{(BE)}$	$u_i(BE) = 0.3 \times BE$	$u_i(BE) = 0.2 \times BE$
	$u_i(SE) = 0.2 (SE)$	$u_i(SE) = 0.1 \times SE$	$u_i(SE) = 0.1 \times SE$
	$u_i(Cue) = step(Cue \geq 0.8)$	$u_i(Cue) = step(Cue \geq 0.9)$	$u_i(Cue) = step(Cue \geq 0.9)$

	$u_i(Ur) = -1$	$u_i(Ur) = -1$	$u_i(Ur) = -0.8$
Young Salaried	$u_i(SU) = 1 \times SU$	$u_i(SU) = 1 \times SU$	$u_i(SU) = SU^2$
	$u_i(SV) = 1 \times SV$	$u_i(SV) = 1/(1 + e^{(-8(SV-0.55))})$	$u_i(SV) = 1/(1 + e^{(-12(SV-0.75))})$
	$u_i(BE) = BE^2$	$u_i(BE) = 0.6 \times BE$	$u_i(BE) = 0.5 \times \sqrt{BE}$
	$u_i(SE) = 1 \times SE$	$u_i(SE) = 1.1 \times SE$	$u_i(SE) = 0.4 \times SE$
	$u_i(Cue) = 1 \times Cue$	$u_i(Cue) = 1 \times Cue$	$u_i(Cue) = \text{step}(Cue \geq 0.8)$
	$u_i(Ur) = 1$	$u_i(Ur) = 0.8$	$u_i(Ur) = 0.6$
Older Hourly	$u_i(SU) = 1/(1 + e^{(-8(SU-0.85))})$	$u_i(SU) = 0.4 \times SU$	$u_i(SU) = 0.4 \times SU$
	$u_i(SV) = 0.6 \sqrt{(SV)}$	$u_i(SV) = 1/(1 + e^{(-10(SV-0.9))})$	$u_i(SV) = 0.2 \times SV$
	$u_i(BE) = 0.4 \times BE$	$u_i(BE) = 0.1 \times BE$	$u_i(BE) = 0$
	$u_i(SE) = 0.5 (SE^2)$	$u_i(SE) = 0.2 (SE)$	$u_i(SE) = 0.1 \times (SE)$
	$u_i(Cue) = \text{step}(Cue \geq 0.6)$	$u_i(Cue) = \text{step}(Cue \geq 0.8)$	$u_i(Cue) = \text{step}(Cue \geq 0.9)$
	$u_i(Ur) = -0.9$	$u_i(Ur) = -1$	$u_i(Ur) = -1$
Old Salaried	$u_i(SU) = 1.2 \times SU^{1.2}$	$u_i(SU) = 0.5 \times SU$	$u_i(SU) = SU^{0.1}$
	$u_i(SV) = 1.5 \sqrt{(SV)}$	$u_i(SV) = 1/(1 + e^{(-10(SV-0.9))})$	$u_i(SV) = 0.2 \times SV$
	$u_i(BE) = 1.5 \times \sqrt{(BE)}$	$u_i(BE) = 0.1 \times BE$	$u_i(BE) = 0.6 \times BE$
	$u_i(SE) = 1.5 (SE)$	$u_i(SE) = 1.1 \times SE$	$u_i(SE) = 0.5 + SE$

	$u_i(Cue) = 1 \times Cue$	$u_i(Cue) = step(Cue \geq 0.8)$	$u_i(Cue) = 1 \times Cue$
	$u_i(Ur) = -0.4$	$u_i(Ur) = -0.6$	$u_i(Ur) = -0.5$

Table 1. Construct functional forms for each disease scenario and persona.

Functional Form	Constructs (Examples)	Personas (Where Observed)	Cognitive Interpretation	Coefficient Interpretation
Concave (Diminishing Returns)	Susceptibility, Benefits	Young Hourly (Mild Cold), Chronic contexts	Beliefs exhibit diminishing marginal sensitivity. Once illness is recognized, additional increases in perceived risk or recovery benefit do not proportionally increase motivation to act. This reflects normalization of mild or familiar symptoms and economic anchoring.	Differences in coefficients reflect relative importance: susceptibility is weighted more than benefits because anticipated worsening carries more urgency than recovery gains, while benefits are attenuated due to skepticism about short-term recovery value.

Linear (Proportional Processing)	Susceptibility, Severity, Self-Efficacy, Cues	Salaried workers (across contexts), some older personas	Beliefs are processed proportionally, with increases in perceived intensity directly translating into decision weight. This reflects environments with fewer structural constraints, allowing illness signals to enter decision-making without suppression.	Coefficients distinguish influence: higher weights (e.g., self-efficacy in salaried workers) reflect autonomy and ability to act, while lower weights (e.g., hourly self-efficacy) indicate that even when processed linearly, the construct has limited impact due to external constraints.
Convex (Accelerating Sensitivity)	Benefits (Preventive reasoning), Susceptibility (some chronic/salaried cases)	Young Salaried (Mild/Chronic contexts)	Beliefs increase slowly at low levels but accelerate at higher levels. This captures preventive reasoning, where individuals increasingly value action (e.g., rest) once benefits or	Coefficients amplify this effect: higher coefficients indicate stronger belief in prevention (e.g., avoiding long-term disruption), while lower coefficients reflect partial skepticism or competing priorities.

			risks become sufficiently salient.	
Sigmoid (Threshold / Legitimacy Gate)	Severity (especially mental health), Susceptibility (chronic/habitua tion cases)	Hourly workers (especially mental health), older workers	Beliefs only influence behavior after crossing a cognitive or social legitimacy threshold. Below this point, symptoms are discounted; above it, they rapidly gain influence. This reflects stigma, endurance norms, and delayed recognition of illness legitimacy.	Threshold placement and steepness encode differences: higher thresholds (e.g., young hourly mental health) indicate stronger suppression and need for extreme symptoms, while lower thresholds (e.g., salaried workers) reflect earlier legitimization of illness.
Step (Binary Trigger)	Cues to Action	Primarily Hourly workers; some chronic/mental contexts	Behavior changes only when a specific condition is met (e.g., explicit approval, medical instruction).	Threshold level reflects reliance on external legitimacy: higher thresholds indicate stronger dependence on formal validation, while lower thresholds reflect

			Internal signals alone are insufficient; external validation is required.	greater responsiveness to moderate cues or experience-based judgment.
--	--	--	---	---

Table 2. Interpretations of functional forms used in HBM construct marginal utility representations.

Illness Scenarios and Input Assumptions

The numerical values assigned to each construct (e.g., susceptibility = 0.4, severity = 0.3) should be interpreted as abstract representations of illness perception, rather than empirically measured quantities. These values are chosen to reflect plausible qualitative distinctions between conditions (e.g., mild vs. severe illness) and to enable comparison across scenarios. For example, lower severity values correspond to conditions perceived as manageable or non-disruptive, while higher values reflect conditions associated with significant impairment. In this sense, the scale is conceptually analogous to disability weights used in the Global Burden of Disease study, which quantify the perceived burden of different health states on a standardized scale (Vos et al., 2020).

Mild Acute Illness (Common Cold)

Mild acute illness represents a short-duration, self-limiting condition with low long-term medical risk such as a cold or mild flu. Susceptibility is set at 0.4 and Severity at 0.3 to reflect that seasonal respiratory infections are often viewed as low-threat "nuisance" illnesses rather than clinical emergencies (BLS, 2023; KFF, 2020). Benefits are set at 0.5, reflecting a moderate belief in recovery through rest, while Self-efficacy at 0.6 indicates a general confidence in self-managing symptoms without professional intervention.

For young hourly workers, susceptibility and severity are processed through concave or convex transformations to reflect the "discounting" of low-level symptoms. Behavioral research on presenteeism suggests that low-threat illness is often marginalized when financial stakes are high (Pichler & Ziebarth, 2017). Barriers are set at 1.0, anchored in labor economics data showing that immediate income loss is a near-total deterrent for low-wage populations who lack a financial "buffer" (Schneider & Harknett, 2021). Benefits are processed conservatively because the immediate utility of a paycheck outweighs the delayed utility of short-term recovery.

For young salaried workers, severity and benefits are processed linearly, reflecting a proportional evaluation of symptoms. Barriers are reduced to 0.5, reflecting a shift from financial

survival to reputational cost (Hemp, 2004). This aligns with evidence that access to paid leave reduces "economic anchoring" and allows internal illness signals to carry greater weight in the utility calculation (Pichler & Ziebarth, 2017; Kumar et al., 2012).

For older hourly workers, susceptibility is flattened to reflect the "normalization" of routine illness. Older workers in physically demanding roles often develop a higher cognitive threshold for what constitutes "sickness," viewing mild discomfort as a baseline functional state (Paggi & Jopp, 2015). Barriers remain high (1.0) due to acute financial vulnerability and the "Social Security anchor," where missing work threatens long-term retirement stability (Boyens et al., 2021).

For older salaried workers, self-efficacy is amplified through positively shifted transformations, reflecting greater autonomy and institutional security. These workers possess higher "occupational self-efficacy" due to tenure (Schyns & von Collani, 2002). Benefits of rest are processed proportionally, as institutional security allows them to prioritize long-term health sustainability over immediate presence (Aronsson & Gustafsson, 2005).

Acute Mental Health Episode (Panic Attack)

In this model panic attacks are used as the representative case of an acute mental health episode. Susceptibility (0.8) and severity (0.9) reflect the high subjective distress and fear of recurrence characteristic of panic disorders (National Health Service [NHS], n.d.). Benefits are set at 0.7 to reflect a belief that withdrawal from the stressor is necessary for stabilization. However, self-efficacy is reduced to 0.3, reflecting the "loss of control" inherent in panic symptoms. Cues are set at 0.2 because psychological distress is often internally processed and lacks the "visible legitimacy" of physical symptoms (Corrigan & Watson, 2002).

For young hourly workers, severity is processed through a high-threshold sigmoid function. This reflects stigma literature indicating that psychological distress often requires "visible impairment" (Corrigan & Watson, 2002). Susceptibility is dampened by "economic

suppression," where the fear of recurrence is cognitively de-weighted to accommodate the necessity of working (Schneider & Harknett, 2021).

For young salaried workers, severity is processed with a lower-threshold sigmoid, reflecting higher mental health literacy and institutional validation of psychological well-being. Benefits are weighted more strongly, consistent with a "preventive orientation" (Kumar et al., 2012). Higher self-efficacy reflects their relative confidence in requesting leave without fear of immediate termination.

For older hourly workers, susceptibility and severity are flattened to reflect "moralized endurance" norms. Research suggests that older generations may interpret acute distress as "general stress" or a character challenge rather than a clinical illness (Paggi & Jopp, 2015). Barriers remain high (0.8–1.0) due to the lack of structural support for mental health in hourly sectors.

For older salaried workers, severity and susceptibility are processed proportionally. Their seniority grants them the "institutional capital" to treat mental distress as a valid health signal. Barriers are reduced, allowing distress to enter the utility calculation meaningfully rather than being filtered out by financial fear (Urban Institute, 2021).

Chronic Illness (Chronic Back Pain)

The chronic illness scenario is parameterized to represent conditions such as chronic back pain. Chronic back pain is characterized by persistence that interferes with daily life (Cleveland Clinic). Susceptibility is set at 0.4 to reflect expected recurrence, while severity is set at 0.7 to reflect familiar but significant impairment. Benefits are low (0.2) because workers often perceive that a single day of rest will not "cure" a multi-year condition. Self-efficacy (0.4) and Cues (0.3) are low due to the lack of acute triggers.

For young hourly workers, susceptibility is flattened through concave transformation to reflect resignation and "habituation." When a condition is permanent, the marginal utility of responding to it decreases (Karanika-Murray & Biron, 2020). Benefits are minimized because

missing work provides no path to resolution, and the 0.8–1.0 barrier of wage loss remains the dominant factor.

For young salaried workers, susceptibility may be processed convexly, reflecting an "optimism bias" where younger workers believe recovery is still possible if they "manage" the condition. However, benefits remain low as the accumulation of work while absent reduces the perceived value of taking time off.

For older hourly workers, susceptibility is flattened to reflect the "normalization" of routine illness. Older workers in physically demanding roles often develop a higher cognitive threshold for what constitutes "sickness," viewing mild discomfort as a baseline functional state (Paggi & Jopp, 2015). Barriers remain high (1.0) due to acute financial vulnerability and long-term economic insecurity (Boyens et al., 2021).

For older salaried workers, severity is processed proportionally and self-efficacy is amplified to reflect strategic self-management. Their autonomy allows them to adjust their work environment (e.g., ergonomic changes or flexible hours) which reflects a high self-efficacy in managing the condition while maintaining employment (Urban Institute, 2021).

One Way Sensitivity Analysis:

A one-way sensitivity analysis was conducted as a diagnostic tool to understand how the model responds to local changes in construct values within each scenario. Because the utility function is additive, changes in each component produce proportional shifts in total utility determined by the functional form of that component. As such, the purpose of this analysis is not to compare global importance across constructs, but to provide intuition about how specific inputs influence decision outcomes near the baseline conditions defined in each scenario.

For each persona and illness context, baseline utility values were first computed using the specified functional transformations of all Health Belief Model (HBM) constructs. Each construct (perceived susceptibility, perceived severity, perceived benefits, self-efficacy, cues to

action, and perceived barriers) was then varied independently while holding all other inputs constant. Specifically, each parameter was increased by 10% of its original value (i.e., $x = x + 0.1x$), with all values bounded within the [0,1] interval. For each perturbation, the total utility was recalculated and compared to the baseline value. The resulting change in utility (ΔU) was recorded, along with whether the predicted behavioral outcome changed (i.e., whether the sign of the utility function crossed zero). This procedure was repeated across all personas and illness contexts.

Results

Scenario Personas by Illness Context and Worker Type

Persona	Mild Acute Illness (Cold)	Mental Health Condition (Panic Attack)	Chronic Illness (Chronic Back Pain)
Young Hourly	Go to Work (-0.35)	Go to Work (-0.35)	Go to Work (-0.63)
Young Salaried	Stay Home (2.95)	Stay Home (3.50)	Stay Home (1.45)
Older Hourly	Go to Work (-0.17)	Go to Work (-0.05)	Go to work (-0.66)
Older Salaried	Stay Home (3.18)	Stay Home (0.70)	Stay Home (1.87)
Values above 0 indicate predicted sick leave utilization; values below 0 indicate predicted continued work attendance.			

Table 3. Summed Utility Values and Predicted Sick Leave Decisions by Persona and Illness

Scenario

Applying the utility-based decision framework described in the Methods section, utility values were computed for each persona across three illness contexts: mild acute illness (common cold), acute mental health episode (panic attack aftermath), and chronic illness (chronic back pain). Behavioral predictions were determined by the sign of total utility, where positive values indicate predicted sick leave utilization since benefits of staying home outweigh the costs and negative values indicate continued work attendance since costs outweigh benefits.

Across all illness contexts, employment structure (hourly versus salaried) produced the most consistent divergence in predicted behavior. Both hourly personas generally exhibited negative utility values, indicating continued work attendance, whereas salaried personas consistently produced positive utility values, indicating sick leave utilization (Table 3). Age introduced additional variation, particularly in chronic illness contexts, where older hourly workers were more likely than younger hourly workers to transition toward positive utility.

Sensitivity Analysis Results

The results of the sensitivity analysis should be interpreted in the context of the model's structure. Because the utility function is additive, the magnitude of change in total utility is directly determined by the functional form and scaling of each component. As a result, the observed sensitivity patterns primarily reflect the specified transformations and coefficients assigned to each construct, rather than providing an independent measure of their real-world importance. Nonetheless, the purpose of this analysis is not to rank constructs globally, but to confirm the internal behavior of the model and to visualize how variation in each input influences utility within the defined scenarios. In this sense, the sensitivity analysis serves as a diagnostic tool, providing intuition about how different

components operate near the baseline conditions and identifying where small changes may meaningfully affect predicted decisions.

LARGEST Sensitivity — Construct with Greatest Impact on Total U						
Persona	Illness	Most Sensitive Construct	Baseline U	Perturbed U (+10%)	ΔU (absolute)	ΔU (%)
Young Hourly	Mild Cold	SU	-0.3492	-0.3337	0.0154	4.42%
Young Hourly	Mental Health	SV	-0.3592	-0.0343	0.3248	90.44%
Young Hourly	Chronic Health	SV	-0.6334	-0.5563	0.0771	12.18%
Young Salaried	Mild Cold	SE	2.95	3.01	0.06	2.03%
Young Salaried	Mental Health	SU	3.4927	3.5727	0.08	2.29%
Young Salaried	Chronic Health	SV	1.498	1.7033	0.2054	13.71%
Older Hourly	Mild Cold	SE	-0.1648	-0.127	0.0378	22.94%
Older Hourly	Mental Health	SV	-0.05	0.1609	0.2109	421.90%
Older Hourly	Chronic Health	SU	-0.66	-0.644	0.016	2.42%
Old Salaried	Mild Cold	SE	3.1819	3.2719	0.09	2.83%

Old Salaried	Mental Health	SV	0.7	0.9109	0.2109	30.14%
Old Salaried	Chronic Health	SE	1.8724	1.9124	0.04	2.14%
SMALLEST Sensitivity — Construct with Least Impact on Total U						
Persona	Illness	Most Sensitive Construct	Baseline U	Perturbed U (+10%)	ΔU (absolute)	ΔU (%)
Young Hourly	Mild Cold	Cue	-0.3492	-0.3492	0	0.00%
Young Hourly	Mental Health	Cue	-0.3592	-0.3592	0	0.00%
Young Hourly	Chronic Health	Cue	-0.6334	-0.6334	0	0.00%
Young Salaried	Mild Cold	SV	2.95	2.98	0.03	1.02%
Young Salaried	Mental Health	Cue	3.4927	3.5127	0.02	0.57%
Young Salaried	Chronic Health	Cue	1.498	1.498	0	0.00%
Older Hourly	Mild Cold	Cue	-0.1648	-0.1648	0	0.00%
Older Hourly	Mental Health	Cue	-0.05	-0.05	0	0.00%
Older Hourly	Chronic Health	BE	-0.66	-0.66	0	0.00%

Old Salaried	Mild Cold	SV	3.1819	3.222	0.0401	1.26%
Old Salaried	Mental Health	Cue	0.7	0.7	0	0.00%
Old Salaried	Chronic Health	SU	1.8724	1.8812	0.0087	0.47%

Table 4. One-way sensitivity analysis: most and least influential HBM constructs by persona and illness context

A one-way sensitivity analysis was conducted by varying each construct independently by +10% of its original value while holding all other inputs constant. Across all personas and illness contexts, changes in perceived barriers produced the largest changes in total utility. In contrast, changes in cues to action frequently resulted in no change in utility, particularly when cue values remained below threshold levels in step-function specifications. Changes in perceived susceptibility, severity, and benefits produced relatively small changes in utility in many cases, particularly for hourly worker conditions. Behavioral reversals (i.e., changes in predicted decision) occurred only in cases where baseline utility values were near zero.

Discussion

This study developed a decision-theoretic model integrating the Health Belief Model to examine sick leave behavior across four worker personas and three illness contexts. The central finding is that employment structure, specifically whether a worker is hourly or salaried, is the strongest and most consistent predictor of behavioral outcomes across all illness types. Hourly workers are predicted to continue working while ill in every scenario modeled, while salaried workers are predicted to take leave. These results suggest that sick leave behavior reflects rational adaptation to structural constraints rather than a failure to recognize or respond to illness. Choices that appear suboptimal from the outside are better understood as consistent with optimization under individual-specific constraints and interpretations of value.

Furthermore, this modeling exercise highlights how different assumptions about cognitive processing (e.g., linear vs. threshold-based responses) can produce meaningfully different behavioral predictions. This underscores the importance of carefully specifying functional forms in decision models, as they implicitly encode assumptions about how individuals interpret and act on information.

Structural barriers suppress illness beliefs: alignment with presenteeism research
Across all illness contexts, the model indicates that the most consistent source of variation in behavior is employment structure rather than illness type or demographic differences (Pichler & Ziebarth, 2017; Schneider & Harknett, 2021; BLS, 2023). These constraints are especially pronounced in hourly and service-sector roles. My model indicates that structural constraints actively suppress the influence of illness-related beliefs within the decision process itself.

The key structural difference between hourly and salaried workers in the model lies in two places: the magnitude of the barrier term and the functional form applied to illness-related beliefs. For hourly workers, barriers enter at values of 0.8–1.0, while susceptibility, severity, and benefits are processed through concave or sigmoid functions that attenuate their contribution. For salaried workers, barriers are lower (0.4–0.6), and illness-related constructs are processed more linearly, allowing them to retain their weight in the utility calculation. The behavioral divergence reflects how both groups share the same input values yet are weighted and transformed differently. This pattern is consistent with behavioral economics research demonstrating that individuals overweight immediate losses relative to delayed benefits (Loewenstein et al., 2001; Camerer et al., 2004). The model reflects how barriers enter as large negative terms representing immediate costs, while illness-related beliefs, concerning future health consequences, are often attenuated through concave or sigmoid transformations.

The importance of operationalization is also highlighted by this model. The functional form assigned to each construct substantially shapes behavior. If susceptibility, benefits, or severity were modeled linearly rather than through concave or sigmoid functions, their

contributions would increase more directly and could produce different behavioral predictions. Transforming the Health Belief Model into a decision model therefore requires not only specifying relevant constructs, but also defining how those constructs are cognitively processed.

Structural Constraints as the Primary Driver of Behavior

Identical illness perceptions can thus produce opposite behavioral outcomes depending on how strongly those perceptions are permitted to enter the utility calculation. While belief formation may vary across groups, these differences appear secondary to how strongly those beliefs are weighted within the decision process.

This finding aligns with existing research showing that workers without paid sick leave are significantly more likely to work while ill because of financial constraints and job insecurity (Pichler & Ziebarth, 2017; Schneider & Harknett, 2021). The present model shows that structural costs do more than constrain options; they determine how strongly illness-related beliefs enter the decision process. Even when susceptibility and benefits are nontrivial, they remain insufficient to outweigh barrier terms in hourly worker conditions.

While age introduces some variation in how illness is interpreted, its effects are comparatively weaker than employment structure. Differences between younger and older workers within the same employment category are relatively small compared to differences between hourly and salaried workers. This is consistent with evidence that structural factors such as wages and access to paid leave remain strong predictors of presenteeism, even when demographic factors such as age are taken into account (Susser & Ziebarth, 2016).

Mild Acute Illness

In the mild acute illness condition, behavior diverges by employment structure. Both hourly worker groups are predicted to continue working, while both salaried worker groups are predicted to take sick leave. For hourly workers, this outcome emerges despite nontrivial contributions from perceived susceptibility and perceived benefits. These constructs capture awareness of future risks, including symptom progression and transmission, as well as the value

of rest for recovery. However, they are attenuated through concave transformations and smaller coefficients, reducing their contribution to utility. In this context, barriers representing direct financial loss dominate the decision.

This pattern is seen in young hourly workers, but it also applies to older hourly workers, who remain similarly constrained. In both cases, illness is recognized but insufficiently weighted to alter behavior. Low self-efficacy and the absence of strong cues to action further reduce the translation of illness-related beliefs into action. Prior research suggests that perceived behavioral control is important in determining whether individuals act on health-related information (Bandura, 1997). Without confidence in the ability to take leave, or without external validation that absence is acceptable, even recognized risks may fail to produce action.

In contrast, both salaried groups are predicted to stay home. Younger salaried workers exhibit positive utility values because illness-related inputs are processed more proportionally, allowing susceptibility and severity to retain their influence. Older salaried workers exhibit even higher utility values, reflecting both reduced barriers and stronger weighting of benefits. These patterns are consistent with research showing that access to paid sick leave reduces presenteeism and enables workers to respond more appropriately to illness (Pichler & Ziebarth, 2017; Schneider & Harknett, 2021). The model extends this literature by demonstrating that lower barriers not only increase the feasibility of staying home, but also change how illness-related inputs enter the decision process.

Mental Health (Panic Attack)

The mental health condition shows the same broad structural split, but with stronger attenuation of illness-related inputs. Both hourly worker groups are predicted to continue working, while both salaried groups are predicted to stay home.

For hourly workers, this result is notable because it occurs despite high input values for perceived severity and susceptibility. These inputs are transformed through sigmoid functions with high thresholds, which substantially reduce their contribution to utility unless they cross a

legitimacy boundary. Compared to the mild illness condition, where some constructs retain partial influence, the mental health condition is characterized by more uniform suppression across belief components. This pattern is consistent with literature on mental health stigma, which suggests that psychological distress is often not recognized as legitimate illness until it reaches a level of visible or externally validated impairment (Corrigan & Watson, 2002; Henderson et al., 2013). The threshold-based transformations capture this dynamic by requiring symptoms to exceed a high activation point before influencing behavior. In this way, the model suggests that individuals may experience substantial distress while still discounting its relevance to the decision to stay home.

Both salaried groups, by contrast, are predicted to stay home because severity is activated at lower thresholds and illness-related inputs are weighted more strongly. Younger salaried workers, in particular, show high utility values, which is consistent with research suggesting that mental health literacy is greater among individuals with more education and stronger institutional support (Jorm, 2000; Reavley & Jorm, 2012). More specifically, for young salaried workers, severity is processed with a lower threshold sigmoid, reflecting a documented generational shift in mental health literacy and decreased stigma among younger cohorts (APA, 2023; Twenge et al., 2019). These workers are significantly more likely to view mental health recovery as a proactive benefit rather than a failure of productivity (Deloitte, 2023). Consequently, their utility calculations reflect a higher self-efficacy in navigating institutional leave policies, as they increasingly expect workplace environments to validate psychological distress as a legitimate health signal (Mind Share Partners, 2021). At the same time, this interpretation reflects modeling assumptions; other factors such as prior experiences or individual/family stigma are not explicitly represented and may also matter.

Chronic Health (Chronic Back Pain)

In the chronic illness condition, structural constraints remain important, but perceived susceptibility becomes more prominent. Both hourly groups remain constrained overall, though older hourly workers are closer to the decision threshold than in the other conditions.

For older hourly workers, susceptibility reflects the expectation that continued physical labor will worsen existing pain. The model indicates the contribution of susceptibility even when current severity is moderate. However, the overall decision remains constrained by barriers, producing utility values that remain near or below zero. This reflects tension between anticipated long-term consequences and immediate structural costs.

This finding is consistent with research on chronic pain adaptation, which shows that individuals often normalize persistent symptoms over time (Eccleston & Crombez, 1999). At the same time, perceived benefits of rest are typically low because short-term absence does not resolve long-term conditions (Turk & Okifuji, 2002). For salaried workers, the model predicts sick leave utilization, though at lower utility values than in acute conditions. This reflects more moderate weighting of illness-related inputs and continued, though reduced, influence of barriers. Differences across age groups may reflect variation in how symptoms are interpreted, with older individuals potentially requiring more concrete indicators of impairment.

This condition also highlights a limitation of the model. The scenario assumes that decisions are made during ongoing discomfort or close to symptom onset. Altering that framing, for example by introducing delays or changing perceived severity, could shift the resulting outcomes. The model therefore illustrates how behavioral predictions depend not only on input values, but also on how decision contexts are cognitively framed.

Sensitivity Analysis

One-way sensitivity analysis demonstrates the local impact of changes in each input value. These effects are governed by the functional form of each respective utility construct. The model is insensitive to variation in cues to action in all scenarios that do not cross the step

function threshold. The question for intervention is not whether cues are present but whether they are strong enough to breach the threshold for a given individual. Perceived barriers exhibit a similar pattern, requiring substantial changes before influencing behavior. This is particularly evident for older hourly workers facing chronic illness, where structural constraints remain dominant.

The model exhibits substantial sensitivity to perceived severity in the mental health incident scenario. For Young Hourly workers, a 10% increase in severity produces a 90.44% change in utility. For Older Hourly workers in the same context, the equivalent perturbation produces a 421.90% shift and is the only instance in the full analysis where utility crosses zero, reversing the predicted behavioral outcome from not seeking care to seeking care. Mental health severity is modeled with sigmoid functions. When baseline severity beliefs sit just below the inflection point, a small increase pushes the function into its steep region and produces disproportionate utility gains. This can be explained as for populations where severity perceptions are suppressed by stigma or limited health literacy, even modest shifts in perceived seriousness could have huge effects. Severity ranked first in the mental health context for three of the four personas. Young Salaried workers are the exception, as their utility is high enough that no single construct moves the needle substantially.

The sensitivity of the model to self-efficacy exhibits noteworthy features for Older Hourly and Old Salaried workers. In the mild cold condition, self-efficacy produces the largest utility shift for Older Hourly workers at 23% and holds second place in the mental health condition at 12%. Old Salaried workers show self-efficacy in the top two constructs across all three illness contexts. The pattern suggests that for these groups, illness beliefs are not the things stopping people from taking sick leave. Severity and susceptibility are present, but the perceived capacity to act on them is not. Older adults and hourly workers often face structural access barriers that internalize as low self-efficacy: difficulty getting time off, uncertainty about navigating care

systems, past experiences of dismissed symptoms. Interventions addressing the logistical dimensions of access are likely to do more for these personas than those aimed at reinforcing illness beliefs alone. Among Young Salaried workers, no single construct dominated consistently, suggesting a more diffuse decision structure.

While the local sensitivity estimates calculated above are a direct consequence of the choice of utility construct functions, they provide important context for the local behavior of the model under different disease, age, and employment scenarios. High utility personas, such as Young Salaried workers with baseline utilities between 1.50 and 3.49, can show large percent changes in utility that are behaviorally meaningless because the decision is nowhere near the threshold. What matters is whether sensitivity falls at a decision-relevant point. It does so most clearly for Older Hourly workers in the mental health condition, where a baseline utility of -0.05 places the group at the margin of care-seeking. A 10% increase in perceived severity is enough to cross it. This is the population the model identifies as most responsive to intervention, and the finding reframes what sensitivity analysis is for in behavioral modeling: not to rank constructs globally, but to locate the specific individuals and contexts where changing beliefs can actually change behavior.

Policy Implications

These findings directly influence how we should design and target sick leave policies. According to the model, hourly employees will continue working across all three illness contexts, even when facing severe conditions like panic attacks and chronic back pain. Because financial and job security constraints are so heavy, health beliefs alone cannot motivate an hourly worker to stay home. Therefore, purely cognitive or behavioral interventions like health education, stigma reduction, or managerial messaging will likely fail if implemented on their own. For this population, structural reform is a strict prerequisite for any other intervention strategy.

First, providing mandated access to paid sick leave will likely produce the largest behavioral impact for hourly workers by directly reducing their primary structural barriers. In the model, these barriers dominate the utility calculation for hourly employees across all illness types. As long as these obstacles remain at or near their ceiling, internal illness-related beliefs remain suppressed.

Second, the effectiveness of cognitive and behavioral interventions depends entirely on how close an individual sits to the decision threshold. In the mental health condition, perceived severity exhibited the highest sensitivity and produced the only behavioral reversal in the entire analysis. This indicates that interventions aimed at lowering legitimacy thresholds can meaningfully increase responsiveness to illness signals. Examples of these interventions include active managerial validation, clear organizational norms, or formal mental health leave policies. In these contexts, the barrier is cognitive as well as structural, meaning employees require their symptoms to be interpreted as valid before they feel they can take action.

Third, increasing self-efficacy and providing clear procedural guidance can help workers translate their beliefs into action, but primarily in less constrained environments. The sensitivity analysis showed that self-efficacy produced its largest utility changes for salaried and older hourly workers. These are groups where structural barriers are already low enough to allow illness signals to enter the decision process. For highly constrained hourly workers, increasing self-efficacy has a limited impact because the threat of income loss continues to suppress all other inputs.

Taken together, these results suggest that effective sick leave policies must align intervention strategies with the worker's specific decision-making environment. Structural interventions are necessary to shift the baseline feasibility for hourly workers. Meanwhile, cognitive and behavioral interventions work best when individuals are already sitting near the decision threshold.

Conclusion

Taken together, these findings suggest that decisions to take or forgo sick leave reflect a rational response to perceived constraints and priorities, rather than a failure to recognize illness. Workers who continue working while ill are not ignoring their health; they are operating within a decision environment in which illness-related concerns are downgraded relative to immediate costs. Conversely, workers who take sick leave are those for whom illness signals are allowed to meaningfully influence behavior, either due to lower barriers, stronger weighting of benefits, or lower legitimacy thresholds.

More broadly, the model demonstrates that behavior depends not only on what individuals believe, but on how those beliefs are transformed within a structured decision system. This suggests that effective interventions must address both structural constraints and cognitive processing, rather than focusing on awareness alone.

Appendix

Baseline Calculations for Utility									
Persona	Illness	u(SU)	u(SV)	u(BE)	u(SE)	u(Cue)	u(Ur)	Total U	Decision
Young Hourly	Mild Cold	0.3162	0.0025	0.2121	0.12	0	-1	-0.3492	Go to work
Young Hourly	Mental Health	0.08	0.3208	0.21	0.03	0	-1	-0.3592	Go to work
Young Hourly	Chronic Health	0.0392	0.0474	0.04	0.04	0	-0.8	-0.6334	Go to work
Young Salaried	Mild Cold	0.4	0.3	0.25	0.6	0.4	1	2.95	Stay home
Young Salaried	Mental Health	0.8	0.9427	0.42	0.33	0.2	0.8	3.4927	Stay home
Young Salaried	Chronic Health	0.16	0.3543	0.2236	0.16	0	0.6	1.498	Stay home
Older Hourly	Mild Cold	0.0266	0.3286	0.2	0.18	0	-0.9	-0.1648	Go to work
Older Hourly	Mental Health	0.32	0.5	0.07	0.06	0	-1	-0.05	Go to work
Older Hourly	Chronic Health	0.16	0.14	0	0.04	0	-1	-0.66	Go to work
Old Salaried	Mild Cold	0.3996	0.8216	1.0607	0.9	0.4	-0.4	3.1819	Stay home
Old Salaried	Mental Health	0.4	0.5	0.07	0.33	0	-0.6	0.7	Stay home
Old Salaried	Chronic Health	0.9124	0.14	0.12	0.9	0.3	-0.5	1.8724	Stay home

Sensitivity Analysis									
Persona	Illness	Construct	Original Value	increased Value (+10%)	Baseline U	Changed U	ΔU (absolute)	ΔU (%)	Sign Change?
Young Hourly	Mild Cold	SU	0.4	0.44	-0.3492	-0.3337	0.0154	4.42%	No
Young Hourly	Mild Cold	SV	0.3	0.33	-0.3492	-0.3478	0.0014	0.40%	No
Young Hourly	Mild Cold	BE	0.5	0.55	-0.3492	-0.3388	0.0104	2.97%	No
Young Hourly	Mild Cold	SE	0.6	0.66	-0.3492	-0.3372	0.012	3.44%	No
Young Hourly	Mild Cold	Cue	0.4	0.44	-0.3492	-0.3492	0	0.00%	No
Young Hourly	Mental Health	SU	0.8	0.88	-0.3592	-0.3512	0.008	2.23%	No
Young Hourly	Mental Health	SV	0.9	0.99	-0.3592	-0.0343	0.3248	90.44%	No
Young Hourly	Mental Health	BE	0.7	0.77	-0.3592	-0.3382	0.021	5.85%	No
Young Hourly	Mental Health	SE	0.3	0.33	-0.3592	-0.3562	0.003	0.84%	No
Young Hourly	Mental Health	Cue	0.2	0.22	-0.3592	-0.3592	0	0.00%	No
Young Hourly	Chronic Health	SU	0.4	0.44	-0.6334	-0.6194	0.014	2.21%	No

Young Hourly	Chronic Health	SV	0.7	0.77	-0.6334	-0.5563	0.0771	12.18%	No
Young Hourly	Chronic Health	BE	0.2	0.22	-0.6334	-0.6294	0.004	0.63%	No
Young Hourly	Chronic Health	SE	0.4	0.44	-0.6334	-0.6294	0.004	0.63%	No
Young Hourly	Chronic Health	Cue	0.3	0.33	-0.6334	-0.6334	0	0.00%	No
Young Salaried	Mild Cold	SU	0.4	0.44	2.95	2.99	0.04	1.36%	No
Young Salaried	Mild Cold	SV	0.3	0.33	2.95	2.98	0.03	1.02%	No
Young Salaried	Mild Cold	BE	0.5	0.55	2.95	3.0025	0.0525	1.78%	No
Young Salaried	Mild Cold	SE	0.6	0.66	2.95	3.01	0.06	2.03%	No
Young Salaried	Mild Cold	Cue	0.4	0.44	2.95	2.99	0.04	1.36%	No
Young Salaried	Mental Health	SU	0.8	0.88	3.4927	3.5727	0.08	2.29%	No
Young Salaried	Mental Health	SV	0.9	0.99	3.4927	3.5213	0.0286	0.82%	No
Young Salaried	Mental Health	BE	0.7	0.77	3.4927	3.5347	0.042	1.20%	No
Young Salaried	Mental Health	SE	0.3	0.33	3.4927	3.5257	0.033	0.94%	No
Young Salaried	Mental Health	Cue	0.2	0.22	3.4927	3.5127	0.02	0.57%	No
Young Salaried	Chronic Health	SU	0.4	0.44	1.498	1.5316	0.0336	2.24%	No

Young Salaried	Chronic Health	SV	0.7	0.77	1.498	1.7033	0.2054	13.71%	No
Young Salaried	Chronic Health	BE	0.2	0.22	1.498	1.5089	0.0109	0.73%	No
Young Salaried	Chronic Health	SE	0.4	0.44	1.498	1.514	0.016	1.07%	No
Young Salaried	Chronic Health	Cue	0.3	0.33	1.498	1.498	0	0.00%	No
Older Hourly	Mild Cold	SU	0.4	0.44	-0.1648	-0.1551	0.0097	5.87%	No
Older Hourly	Mild Cold	SV	0.3	0.33	-0.1648	-0.1487	0.016	9.73%	No
Older Hourly	Mild Cold	BE	0.5	0.55	-0.1648	-0.1448	0.02	12.14%	No
Older Hourly	Mild Cold	SE	0.6	0.66	-0.1648	-0.127	0.0378	22.94%	No
Older Hourly	Mild Cold	Cue	0.4	0.44	-0.1648	-0.1648	0	0.00%	No
Older Hourly	Mental Health	SU	0.8	0.88	-0.05	-0.018	0.032	64.00%	No
Older Hourly	Mental Health	SV	0.9	0.99	-0.05	0.1609	0.2109	421.90%	YES ⚠
Older Hourly	Mental Health	BE	0.7	0.77	-0.05	-0.043	0.007	14.00%	No
Older Hourly	Mental Health	SE	0.3	0.33	-0.05	-0.044	0.006	12.00%	No
Older Hourly	Mental Health	Cue	0.2	0.22	-0.05	-0.05	0	0.00%	No
Older Hourly	Chronic Health	SU	0.4	0.44	-0.66	-0.644	0.016	2.42%	No

Older Hourly	Chronic Health	SV	0.7	0.77	-0.66	-0.646	0.014	2.12%	No
Older Hourly	Chronic Health	BE	0.2	0.22	-0.66	-0.66	0	0.00%	No
Older Hourly	Chronic Health	SE	0.4	0.44	-0.66	-0.656	0.004	0.61%	No
Older Hourly	Chronic Health	Cue	0.3	0.33	-0.66	-0.66	0	0.00%	No
Old Salaried	Mild Cold	SU	0.4	0.44	3.1819	3.2303	0.0484	1.52%	No
Old Salaried	Mild Cold	SV	0.3	0.33	3.1819	3.222	0.0401	1.26%	No
Old Salaried	Mild Cold	BE	0.5	0.55	3.1819	3.2336	0.0518	1.63%	No
Old Salaried	Mild Cold	SE	0.6	0.66	3.1819	3.2719	0.09	2.83%	No
Old Salaried	Mild Cold	Cue	0.4	0.44	3.1819	3.2219	0.04	1.26%	No
Old Salaried	Mental Health	SU	0.8	0.88	0.7	0.74	0.04	5.71%	No
Old Salaried	Mental Health	SV	0.9	0.99	0.7	0.9109	0.2109	30.14%	No
Old Salaried	Mental Health	BE	0.7	0.77	0.7	0.707	0.007	1.00%	No
Old Salaried	Mental Health	SE	0.3	0.33	0.7	0.733	0.033	4.71%	No
Old Salaried	Mental Health	Cue	0.2	0.22	0.7	0.7	0	0.00%	No
Old Salaried	Chronic Health	SU	0.4	0.44	1.8724	1.8812	0.0087	0.47%	No

Old Salaried	Chronic Health	SV	0.7	0.77	1.8724	1.8864	0.014	0.75%	No
Old Salaried	Chronic Health	BE	0.2	0.22	1.8724	1.8844	0.012	0.64%	No
Old Salaried	Chronic Health	SE	0.4	0.44	1.8724	1.9124	0.04	2.14%	No
Old Salaried	Chronic Health	Cue	0.3	0.33	1.8724	1.9024	0.03	1.60%	No

References

- Aronsson, G., & Gustafsson, K. (2005). Sickness presenteeism: Prevalence, attendance-pressure factors, and an outline of a model for research. *Journal of Occupational and Environmental Medicine*, 47(9), 958–966.
- Bandura, A. (1997). *Self-efficacy: The exercise of control*. W. H. Freeman.
- Becker, M.H. (1974) The Health Belief Model and Personal Health Behavior. *Health Education Monographs*, 2, 324- 508. <http://dx.doi.org/10.1177/109019817400200407>
- Boyens, C., Karpman, M., & Smalligan, J. (2021). Access to paid leave is lowest among workers with the greatest needs. Urban Institute. <https://www.urban.org/research/publication/access-paid-leave-lowest-among-workers-gr-eatest-needs>
- Bureau of Labor Statistics. (2023). Employee benefits in the United States, March 2023. U.S. Department of Labor. <https://www.bls.gov/ebs/publications/employee-benefits-in-the-united-states-march-2023.htm>
- Camerer, C. F., Loewenstein, G., & Rabin, M. (Eds.). (2004). *Advances in behavioral economics*. Princeton University Press.
- Centers for Disease Control and Prevention. (2020). Coronavirus disease 2019 (COVID-19): Businesses and workplaces. <https://stacks.cdc.gov/view/cdc/88409>

Champion, V. L., & Skinner, C. S. (2008). The health belief model. In K. Glanz, B. K. Rimer, & K. Viswanath (Eds.), *Health behavior and health education: Theory, research, and practice* (4th ed., pp. 45–65). Jossey-Bass.

Cleveland Clinic. (n.d.). Chronic pain: What it is, symptoms, treatment & causes.

<https://my.clevelandclinic.org/health/diseases/4798-chronic-pain>

Corrigan, P. W., & Watson, A. C. (2002). Understanding the impact of stigma on people with mental illness. *World Psychiatry*, 1(1), 16–20.

Eccleston, C., & Crombez, G. (1999). Pain demands attention: A cognitive-affective model of the interruptive function of pain. *Psychological Bulletin*, 125(3), 356–366.

<https://doi.org/10.1037/0033-2909.125.3.356>

Eurofound. (2012). Fifth European Working Conditions Survey: Overview report. Publications Office of the European Union.

<https://www.eurofound.europa.eu/en/publications/2012/fifth-european-working-conditions-survey-overview-report>

Gigerenzer, G., & Selten, R. (Eds.). (2001). *Bounded rationality: The adaptive toolbox*. MIT Press.

Hemp, P. (2004). Presenteeism: At work—but out of it. *Harvard Business Review*, 82(10), 49–58.

Kaiser Family Foundation. (2020, March 10). Paid sick leave is much less common for lower-wage workers in private industry.

<https://www.kff.org/covid-19/paid-sick-leave-is-much-less-common-for-lower-wage-workers-in-private-industry/>

Karanika-Murray, M., & Biron, C. (2020). The health-performance framework of presenteeism:

Towards understanding an adaptive behavior. *Human Relations*, 73(2), 242–261.

<https://doi.org/10.1177/0018726719827081>

Kumar, S., Quinn, S. C., Kim, K. H., Daniel, L. H., & Freimuth, V. S. (2012). The impact of

workplace policies and other social factors on self-reported influenza-like illness

incidence during the 2009 H1N1 pandemic. *American Journal of Public Health*, 102(1),

134–140. <https://doi.org/10.2105/AJPH.2011.300307>

Loewenstein, G. F., Weber, E. U., Hsee, C. K., & Welch, N. (2001). Risk as feelings.

Psychological Bulletin, 127(2), 267–286. <https://doi.org/10.1037/0033-2909.127.2.267>

Merrell, W. N., Choi, S., & Ackerman, J. M. (2024). When and why people conceal infectious

disease. *Psychological Science*, 35(3). <https://doi.org/10.1177/09567976231221990>

National Health Service. (n.d.). Panic disorder.

<https://www.nhs.uk/mental-health/conditions/panic-disorder/>

Organisation for Economic Co-operation and Development. (2019). Policy responses to new

forms of work. OECD Publishing.

https://www.oecd.org/en/publications/policy-responses-to-new-forms-of-work_0763f1b7-en.html

Paggi, M. E., & Jopp, D. S. (2015). Outcomes of occupational self-efficacy in older workers. *The International Journal of Aging and Human Development*, 80(4), 357–378.

<https://doi.org/10.1177/0091415015607640>

Pichler, S., & Ziebarth, N. R. (2017). The pros and cons of sick pay schemes: Testing for contagious presenteeism and shirking behavior. *Journal of Public Economics*, 156, 14–33. <https://doi.org/10.1016/j.jpubeco.2017.07.003>

Rosenstock, I. M. (1974). Historical origins of the health belief model. *Health Education Monographs*, 2(4), 328–335. <https://doi.org/10.1177/109019817400200403>

Schneider, D., & Harknett, K. (2021). Hard times: Routine schedule unpredictability and material hardship among service sector workers. *Social Forces*, 99(4), 1682–1709. <https://doi.org/10.1093/sf/soaa079>

Schyns, B., & von Collani, G. (2002). A new occupational self-efficacy scale and its relation to personality constructs and organizational variables. *European Journal of Work and Organizational Psychology*, 11(2), 219–241. <https://doi.org/10.1080/13594320244000148>

Simon, H. A. (1955). A behavioral model of rational choice. *The Quarterly Journal of Economics*, 69(1), 99–118. <https://doi.org/10.2307/1884852>

Susser, P., & Ziebarth, N. R. (2016). Profiling the U.S. sick leave landscape: Presenteeism among females. *Health Services Research*, 51(6), 2305–2317.

<https://doi.org/10.1111/1475-6773.12471>

Thaler, R. H., & Sunstein, C. R. (2008). *Nudge: Improving decisions about health, wealth, and happiness*. Yale University Press.

Turk, D. C., & Okifuji, A. (2002). Psychological factors in chronic pain: Evolution and revolution. *Journal of Consulting and Clinical Psychology*, 70(3), 678–690.

<https://doi.org/10.1037/0022-006X.70.3.678>

Urban Institute. (2021). Access to paid leave is lowest among workers with the greatest needs.

<https://www.urban.org/research/publication/access-paid-leave-lowest-among-workers-gr-eatest-needs>

Vos, T., Lim, S. S., Abbafati, C., Abbas, K. M., Abbasi, M., Abbasifard, M., ... Murray, C. J. L. (2020). Global burden of 369 diseases and injuries in 204 countries and territories, 1990–2019: A systematic analysis for the Global Burden of Disease Study 2019. *The Lancet*, 396(10258), 1204–1222. [https://doi.org/10.1016/S0140-6736\(20\)30925-9](https://doi.org/10.1016/S0140-6736(20)30925-9)